REVIEW

After the last Ozempic injection

REFERENCES	STYLE	MADE WITH	VERIFICATION	COMPILED
31 verified	Popsci	Grounded v0.5.1	Crossref · claims	4 Sep 2026

What happens when you stop taking Ozempic? Trials have followed people beyond their final dose, revealing which changes last and which begin to fade.

Rubino and colleagues gave everyone in their withdrawal trial the same beginning: twenty weeks of semaglutide, followed by a random assignment to keep taking it or switch to placebo.[1] Over the next 48 weeks, the continuing group lost more weight, while the switch group regained it.[1] The average changes were -7.9% with continued treatment and +6.9% after switching.[1]

That experiment puts a concrete question behind the familiar one about stopping. How much of a treatment's benefit survives when treatment ends? The randomized withdrawal trial found weight regain after the placebo switch.[1] A longer extension also found regain after treatment ended.[2] Together, they make weight the clearest part of the answer. Appetite, blood sugar and the possibilities for maintaining a loss need their own evidence.

Ozempic is injectable semaglutide, introduced for type 2 diabetes.[3] The STEP 1 withdrawal extension studied semaglutide 2.4 mg in adults without diabetes.[2] If you are asking about your own future, that distinction matters: the studies describe particular people taking particular regimens, rather than every situation in which a prescription ends.

01 THE FIRST CHANGE IS GRADUAL

The molecule does not disappear as soon as an injection is missed. Yang and Yang's review of human drug-handling studies describes a long elimination half-life that supports weekly injections.[3] The half-life describes how slowly the drug clears.[3] Knudsen and Lau describe how semaglutide's reversible binding to albumin prolongs its action.[4]

These measurements establish gradual elimination. They do not give you an exact date for a returning appetite. To see what might change as exposure falls, researchers first had to measure what the drug does while people take it.

Blundell's team offered volunteers meals and snacks they could eat freely.[5] During semaglutide treatment, they ate about a quarter less across those meals than during placebo treatment.[5] In a later experiment, Friedrichsen and colleagues found less hunger, fewer or weaker cravings and better control of eating with the obesity-treatment dose.[6] Researchers testing oral semaglutide also found reduced food intake and appetite in adults with obesity.[7] A separate diabetes study found greater fullness after a fat-rich breakfast.[8] A review of randomized trials of GLP-1 drugs, the class that includes semaglutide, reached the same broad conclusion about appetite suppression.[9]

Here, the distinction between measurement and inference matters. Blundell measured food intake during treatment.[5] Friedrichsen measured appetite during treatment.[6] Their results make a return of appetite pressure after stopping plausible; they do not describe a measured day-by-day hunger curve after the final injection.

Nor does a simple story about food sitting in the stomach explain everything. At the later treatment assessment, Friedrichsen's group found no evidence of delayed gastric emptying using paracetamol absorption as an indirect test.[6] Gabe's oral-semaglutide experiment reported the same late-treatment null finding.[7] Neither result rules out earlier or individual gastric effects.

02 THEN THE PATHS SEPARATE

During active treatment, the weight findings are consistent. In STEP 1, Wilding and colleagues reported a mean weight change of -14.9% with semaglutide.[10] Garvey's STEP 5 team reported a mean weight change of -15.2% with semaglutide at week 104.[11] Moiz and colleagues found substantial long-term weight loss relative to placebo.[12] Bergmann's review describes similar losses across the STEP trials.[13] Gudzune and Kushner's broader medication review also found a substantial semaglutide advantage.[14]

Rubino's withdrawal experiment asked whether that benefit needed continued treatment. Both randomized groups kept their lifestyle intervention, yet the people switched to placebo gained weight while those continuing semaglutide lost more.[1] Only people who had reached the maintenance dose were randomized.[1] Random assignment makes this a particularly useful comparison for separating withdrawal from continued treatment.

The main weight result is visible in Figure 1: continued treatment and withdrawal send the averages in opposite directions.

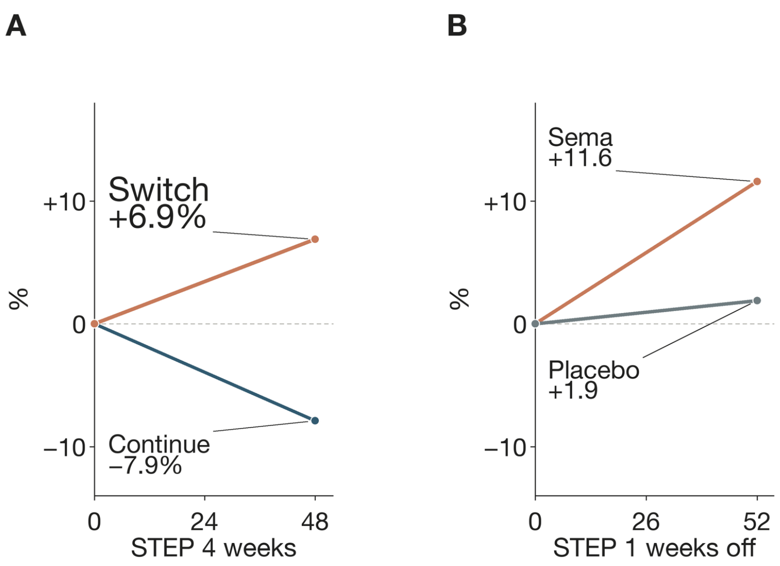

Figure 1. Weight rises after treatment ends. Horizontal axes: follow-up weeks. Vertical axes: percentage change from randomization (A) or percentage-point change from original baseline (B). The withdrawal trial compared continuation with a placebo switch.[1] In the STEP 1 extension, medication and lifestyle support ended.[2] Lines connect endpoints, not measured weekly paths. No uncertainty intervals are plotted. "Sema" means semaglutide.

Wilding's STEP 1 extension followed participants for another year after treatment ended.[2] Former semaglutide recipients regained about two-thirds of their earlier loss, but still averaged less than their starting weight.[2] At follow-up, 48.2% retained at least 5% weight loss.[2] The word "regain" therefore describes a direction, not a claim that everyone returned to the beginning.

03 AVERAGES ACQUIRE A MISLEADING PRECISION

West and colleagues assembled cessation studies across weight-management medicines.[15] For newer incretin medicines, including semaglutide and tirzepatide, their model estimated average regain of 0.8 kg per month.[15] That sounds temptingly like a personal timetable. It is a study-level average.

The researchers had no more than a year of post-cessation observation for the newer medicines; the later trajectory was projected.[15] A line continuing into the future is doing a different job from a measurement collected there.

Other syntheses reinforce the direction while showing why one number cannot fit every setting. Berg's team found that larger initial losses were followed by larger regain.[16] Tzang and colleagues found regain across randomized obesity trials, with very large differences between studies.[17] Quarenghi's review likewise described regain after interruption of several related medicines.[18] These syntheses should be read as summaries of studies, not added together as independent replications.

Figure 2 shows three published estimates with their different scopes. The table keeps the individual study clocks separate.

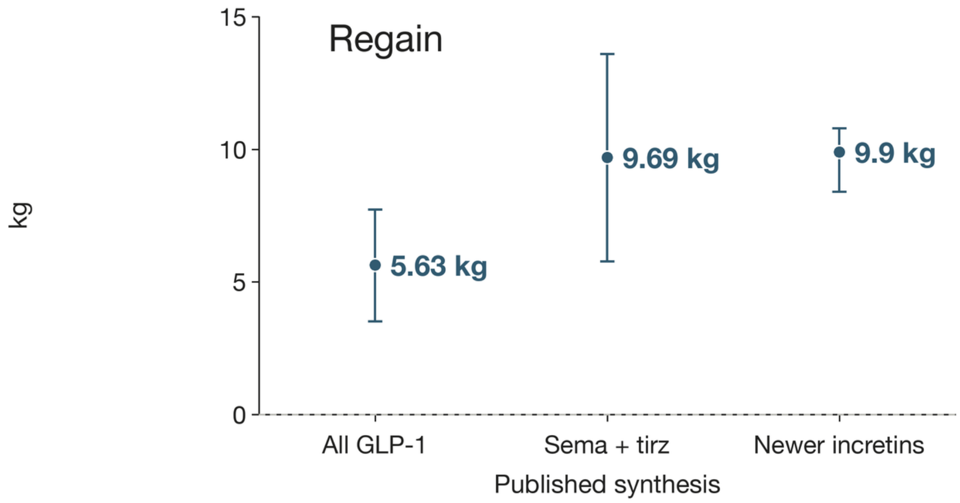

Figure 2. Regain estimates depend on the question. Horizontal categories identify syntheses; vertical position gives kilograms regained. Left to right: Tzang, Berg, West. Tzang pooled GLP-1 cessation in obesity trials.[17] Berg reported regain after semaglutide/tirzepatide cessation.[16] West modeled regain within the first year after newer-incretin cessation.[15] Whiskers show reported 95% confidence intervals. Drugs, follow-up and analyses differ.

EVIDENCE WINDOW	RESULT AFTER STOPPING	BOUNDARY
Withdrawal trial	+6.9% over 48 weeks[1]	Lifestyle intervention continued; maintenance-dose completers
STEP 1 extension	11.6 percentage points over 1 year[2]	Medication and lifestyle intervention both ended
Newer-incretin synthesis	Monthly regain: 0.8 kg; interval 0.7–0.9[15]	Study-level estimate; observation no longer than 12 months
PCOS cohort	About one-third of prior loss regained over two years[19]	Small uncontrolled cohort continuing metformin

04 BLOOD TESTS FOLLOW THE SCALE

Kosiborod and colleagues examined the STEP trials beyond body weight.[20] Waist size, blood pressure, fasting glucose, insulin and lipids improved during STEP 4's common treatment period, then deteriorated after the placebo switch while remaining improved with continued semaglutide.[20] In STEP 1, several benefits also moved back toward baseline after withdrawal, although some remained at the end of follow-up.[2] Wilkinson's team reported a similar pattern in a calculated diabetes-risk score.[21] A calculated risk score is not an observed diagnosis.

Tzang's pooled analysis also tracked glycated haemoglobin, HbA1c.[17] It rose after GLP-1 treatment ended, with a larger average rise in the diabetes studies than in the obesity studies.[17] Variation between studies was especially large for that diabetes estimate.[17]

These findings support moderate confidence that several metabolic improvements commonly diminish after cessation. They cannot settle the fate of every clinical benefit. Lincoff's SELECT trial found fewer major cardiovascular events with semaglutide during randomized treatment assignment.[22] That comparison does not directly test how long protection lasts after stopping. A change in a blood test cannot answer the missing event question.

05 THE PRESCRIPTION DOES NOT ALWAYS END NEATLY

Rodriguez and colleagues looked at routine US care and found that stopping and restarting were both common.[23] Their records tell a less orderly story than a scheduled trial withdrawal. In Gasoyan's chart review, cost or insurance problems were the most frequently recorded reason for stopping; side effects and shortages were also recorded.[24]

Figure 3 separates three questions that can otherwise become blurred: who stopped, who stayed on treatment, and who restarted.

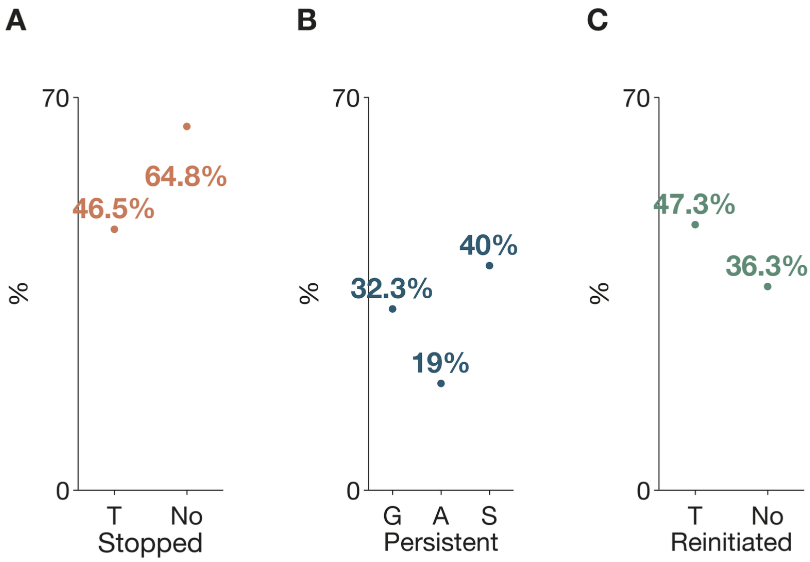

Figure 3. Stopping and restarting have different denominators. Vertical axes: percentages. Horizontal categories: diabetes status (A/C), medication cohorts (B). A shows stopping and C shows restarting, with "T" for type 2 diabetes and "No" for no type 2 diabetes. Rodriguez measured one-year discontinuation and one-year reinitiation after stopping.[23] B: persistence. "G": Gleason's GLP-1 cohort; "A": Gasoyan's all-medication cohort; "S": its semaglutide subgroup. Gleason measured one-year GLP-1 persistence.[25] Gasoyan reported one-year persistence across anti-obesity medications and for semaglutide.[26] Compare each panel within its own study and outcome definition.

Patients with longer medication coverage also lost more weight on average in Gasoyan's clinical cohorts.[27,28] The discontinuation analysis used a retrospective cohort.[27] The earlier semaglutide/liraglutide comparison also used a retrospective cohort.[28] Access, dose and the characteristics of people who continue can contribute to the gap; the cohorts cannot assign the entire difference to stopping alone.

06 SOME PEOPLE KEEP MORE OF THE LOSS

Must the average be everyone's outcome? Jensen and colleagues tested a promising complication: supervised exercise alongside liraglutide, a related GLP-1 medicine.[29] After treatment ended, people previously assigned to the combination maintained weight and fat loss better than those assigned to liraglutide alone, although regain was not eliminated.[29] The drug was different, and follow-up was incomplete.[29] It is an encouraging clue for semaglutide, rather than a demonstrated replacement for it.

Jensterle's team followed a small group of women with polycystic ovary syndrome who continued metformin after semaglutide ended.[19] They regained only about a third of their prior loss over two years, but several metabolic improvements returned to baseline.[19] The study had no control group.[19] It therefore cannot establish that metformin caused the better weight maintenance.

McKenzie and Athinarayanan also reported maintained weight in selected patients who stopped GLP-1 treatment while continuing a carbohydrate-restricted telemedicine program.[30] That comparison used a retrospective cohort, too.[30]

The larger picture is less reassuring than either small cohort alone. West's synthesis found no evidence that post-cessation behavioral support changed regain rates across the included studies.[15] Dash's review likewise calls for randomized testing of proposed lifestyle strategies.[31] Different interventions and selected populations leave room for benefit, but no broadly reliable semaglutide transition protocol emerges from these results.

07 WHAT THE NEXT TRIAL NEEDS TO FOLLOW

Return to Rubino's two groups. Their divergent paths give strong evidence that continuing semaglutide maintains weight benefit better than withdrawing it among people who reach the study dose.[1] The extension found regain and loss of several metabolic improvements.[2] The randomized metabolic analysis found deterioration after the placebo switch.[20] Together, these results support moderate confidence in a common pattern of regain and diminishing benefits.

The next useful experiment would follow people from their final semaglutide dose, randomly compare realistic maintenance strategies, and keep measuring hunger, weight and blood sugar together. It would test which people can preserve which benefits, for how long. That is the unanswered question behind the last injection.

Scope and methods. Narrative review; searches rerun on 4 September 2026 in PubMed, with backward and forward OpenAlex citation chasing. Included peer-reviewed adult human studies of semaglutide and explicitly identified related-drug evidence; excluded anecdotes and preprints. The saved access manifest contains twelve authenticated full texts, eleven cited here; remaining cited material was abstract-level, including the direct STEP 4 report. Outcome confidence reflects design, bias, inconsistency, indirectness and imprecision; it is separate from whether a quotation supports a sentence. Related trial reports are grouped, and overlapping reviews are not counted as independent confirmation.

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